Optimization Projects

Energy System Modeling and Optimization (VU)



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1 W.I.N.E.



Figure 1: Sika deer (*Cervus nippon*) in Nara, Japan

The Hochalm valley is one of the last larger wildlife retreats in the region -– and it is falling apart. Over the decades, a growing city, industrial areas and a motorway (and some natural obstacles such as a mountain ridge and a river) have cut the landscape into nine isolated habitats (H1–H9). Small, isolated populations lose genetic diversity, and animals that try to cross a road do not always survive.

The regional government has launched the programme **W.I.N.E.** (*Wildlife Interaction and Network Efficiency*): protected **wildlife corridors** should reconnect the habitats. Based on feasibility studies, 20 corridors can be built (Figure 2). Each corridor connects exactly two habitats and can be used by animals in both directions. Where a corridor crosses a road, a green bridge or an underpass is needed – this is already included in the construction costs, but each crossing remains a risk for animals and a political issue. A fully connected valley would be the wildest dream of every conservationist – but the budget is rarely that generous. You have been hired as optimization expert to find out which corridors should be built.

Data. Each habitat has a *quality* q (1 = poor, 10 = excellent) and an estimated number of animals n. The *ecological value* of a habitat is defined as $q \cdot n$. Habitat H1 is the national park. It is part of the network in any case (reference Table 1).

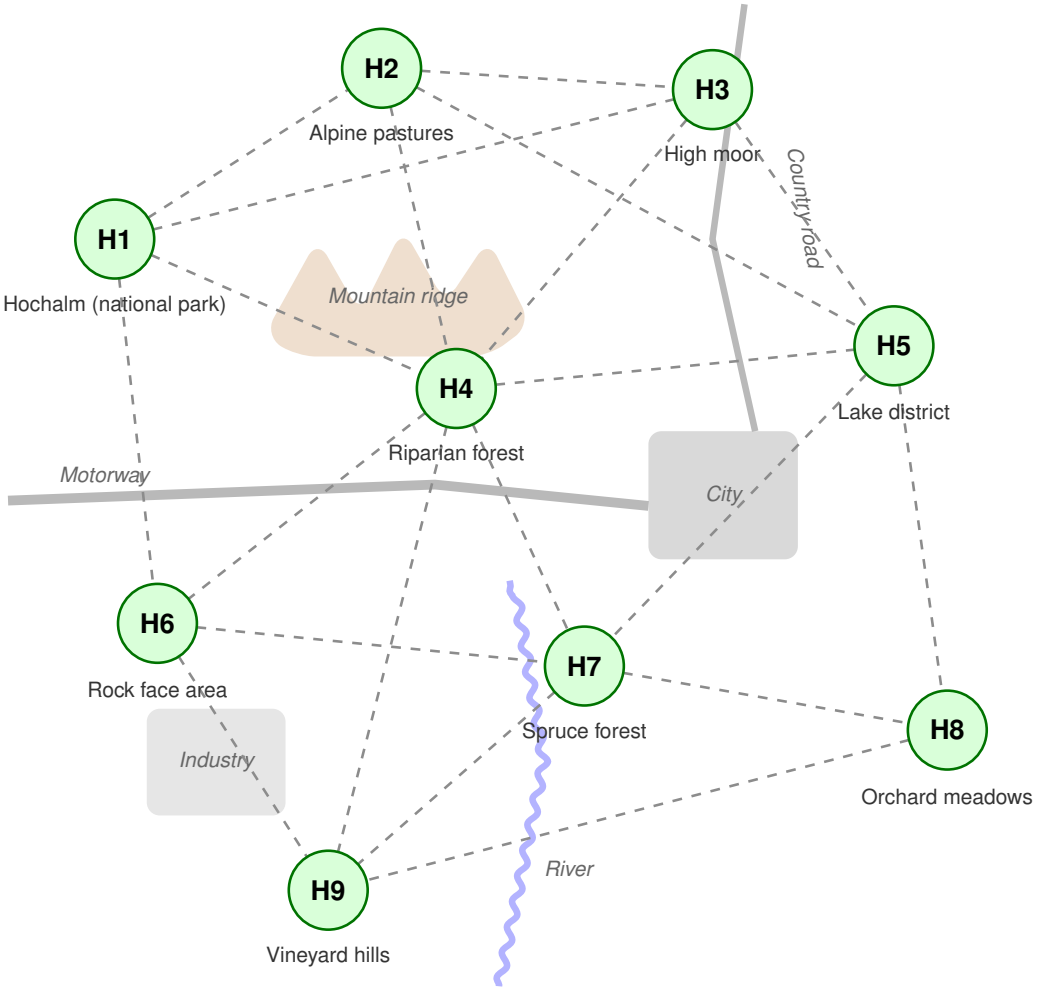


Figure 2: Schematic map of the region (not to scale)

Habitat	Name	Quality q	Animals n
H1	Hochalm (national park)	9	120
H2	Alpine pastures	7	80
H3	High moor	8	40
H4	Riparian forest	6	150
H5	Lake district	8	60
H6	Rock face area	7	90
H7	Spruce forest	5	200
H8	Orchard meadows	4	110
H9	Vineyard hills	3	70

Table 1: Habitat overview

Corridor	Length (km)	Road crossings (#)	Costs (mio. €)
H1–H2	7.9	0	3.1
H1–H3	15.2	0	6.5
H1–H4	9.6	1	5.7
H1–H6	10.2	1	5.4
H2–H3	8.4	0	3.4
H2–H4	8.7	1	4.9
H2–H5	15.8	0	6.7
H3–H4	8.8	2	6.6
H3–H5	5.9	1	3.9
H4–H5	12.7	2	7.9
H4–H6	9.1	0	3.7
H4–H7	4.9	1	3.4
H4–H9	10.5	2	6.4
H5–H7	10.3	1	5.8
H5–H8	8.2	0	3.4
H6–H7	11.9	2	6.9
H6–H9	7.7	1	4.4
H7–H8	10.6	0	4.5
H7–H9	6.9	0	2.9
H8–H9	14.6	1	7.2

Table 2: Corridor overview

Remark: The corridors in the map are schematic. Only the table values are relevant. Corridors that are not listed cannot be built (e.g. due to mountains or buildings).

Formulate a **linear** mixed-integer optimization model for the following tasks, implement it in GAMS and answer the questions. The tasks build on each other.

(1) Connecting all habitats

Two habitats are *connected* if animals can walk from one to the other using only built corridors (also via other habitats in between). Find the cheapest set of corridors such that **all habitats are connected** to each other.

a) Which corridors should be built to connect all habitats at minimum cost? State the total costs, the number of corridors and the number of road crossings.

b) *Redundancy:* If a corridor is blocked (e.g. by a landslide), the network should still be useful. Therefore, every habitat has to be attached to **at least two corridors** (all habitats still have to be connected). Which corridors are built now? How much more expensive is this network compared to a)? Briefly discuss what changes in the structure of the network.

(2) Limited budget

The budget for W.I.N.E. is limited to **16 mio. €**, and the government allows at most **one road crossing** in total. Therefore, not all habitats can be connected any more. The national park (H1) is always part of

the network. Now, you may choose which habitats should be connected, such that the **ecological value of all connected habitats** ($\sum q \cdot n$) is maximized. All connected habitats must be connected with each other.

c) Which habitats are connected and which corridors are built? State the achieved ecological value and the costs. How does the result change if the limit on road crossings is removed? Briefly interpret.

d) The government considers different budgets. Solve the model of c) (with at most one road crossing) for budgets of **12, 16, 20 and 24 mio. €** and summarize the results in a table (budget, connected habitats, ecological value, costs). Is the optimal network for a higher budget always an extension of the network for a lower budget? How much additional value do you get per additional mio. € of budget? Explain your observations.

(3) Challenge: Species with different needs

Not all animals use all corridors. Two protected species have special requirements: each species only accepts corridors up to a maximum length, and it only needs to be able to reach *its own* habitats.

Species	Max. corridor length	Habitats
Fire salamander	8 km	H4, H6, H7, H9
Lynx	10 km	H3, H5, H7, H8

e) Extend the model of a): All habitats still have to be connected, and in addition, each species must be able to reach all its habitats using only built corridors that are not longer than its maximum corridor length (other habitats may be used as intermediate stops). Which corridors are built now, and what are the total costs? Compare with a) and explain the differences.

2 Chasing the red flags



Figure 3: Red-white-red trail markers, commonly seen in the Austrian Alps

Four days, three mountain huts and one long line of red-white-red trail markers: the *Red Flag Trail* leads you from the valley over a high ridge and back down again. Your task is to prepare the trip as well as possible. You have to pack the **backpack** (Part 1) and to decide what to **wear** on which day (Part 2). Both parts are connected: whatever you pack in Part 1 takes away volume, weight and budget from your clothing in Part 2.

Most of your gear comes from two sources: the **attic of your grandparents** (old, free of charge, but usually heavy and bulky) and the **outdoor shop** (modern, light and compact, but it costs money). Both are listed in Table 4 and Table 5.

The trail

The trail starts in the valley at 700 m and consists of four stages (Figure 4). Stages 1 to 3 end in a mountain hut (A, B, C), the last stage ends in the valley. The weather forecast is known for each stage (Table 3).

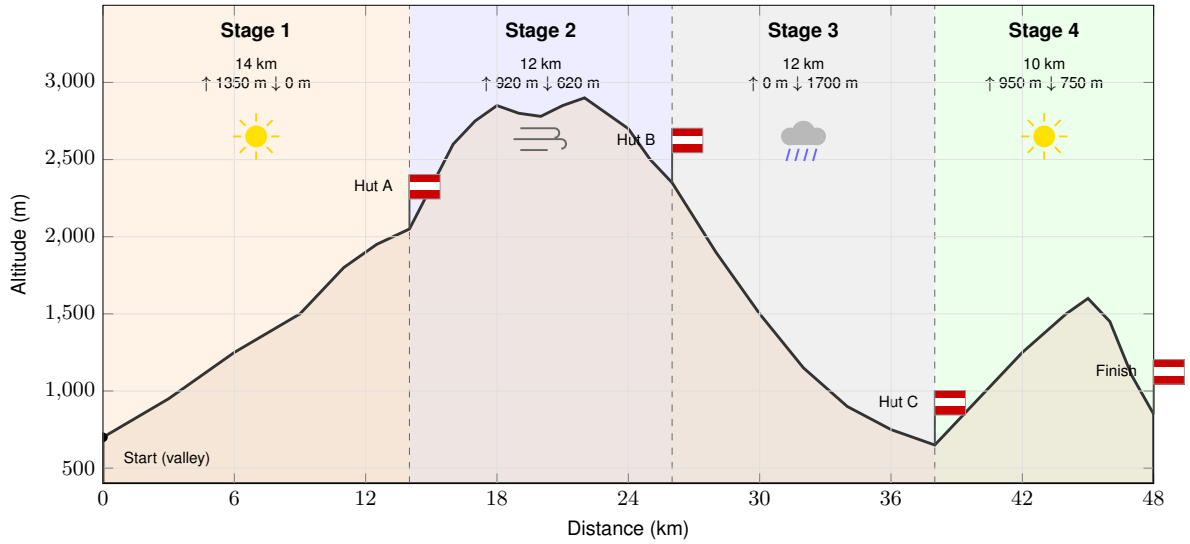


Figure 4: Elevation profile of the Red Flag Trail with the weather forecast of each stage

Stage	Distance (km)	Ascent (m)	Descent (m)	Highest point (m)	Valley temp. at 700 m (°C)	Wind (km/h)	Rain (mm/h)
1	14	1350	0	2050	24	10	0
2	12	920	620	2900	16	55	0
3	12	0	1700	2350	14	25	4
4	10	950	750	1600	21	12	0

Table 3: Stage data and weather forecast. Wind and rain refer to the high part of the stage.

Effort. The effort of stage d (in stamina points) is

$$E_d = \text{distance}_d + 0.02 \cdot \text{ascent}_d + 0.01 \cdot \text{descent}_d,$$

with distance in km and ascent/descent in m.

Rules of the mountains. The following rules of thumb help you to judge which clothing is useful on which stage:

- *Sweating:* The more ascent a stage has, the more you sweat. Then breathable clothing is important, and thick layers are a burden. Descents and flat parts are less critical.
- *Temperature:* With every 100 m of altitude gain the temperature drops by about 0.65 °C with respect to the temperature at the valley. The relevant temperature of a stage is the one at its highest point.
- *Wind:* Wind is strongest on exposed ridges at high altitude. As a rule of thumb, it feels about 1 °C colder per 10 km/h of wind speed. Wind-proof clothing is then valuable.
- *Rain:* Rain is only a problem if it actually rains. Heavy rain (several mm/h) soaks poor clothing and cools you down quickly.
- *Sun:* On sunny days, the UV radiation increases with altitude. A hat and sun protection help, and thick insulation is of no use (it is even annoying) when it is warm.

(1) Packing the backpack

Table 4 lists the available equipment. Each item has a weight, a volume, a price and a *benefit* (in benefit points, BP). For every group marked as necessary, **exactly one item** has to be taken. "Poles" are optional, but you can only use one set: either the wooden or the carbon poles (or none). The other optional items can be chosen freely. The equipment may use at most **10 litres** of the backpack volume.

Group	Item	Origin	Weight (kg)	Volume (l)	Price (€)	Benefit (BP)
Sleeping	Wool blanket sack	old	1.4	6	0	5
Sleeping	Silk sleeping liner	new	0.18	0.8	40	5
Light	Flashlight	old	0.35	0.6	0	3
Light	Headlamp	new	0.09	0.15	35	5
First aid	First-aid tin	old	0.85	1.8	0	5
First aid	Compact first-aid kit	new	0.3	0.6	25	5
Navigation	Paper map and compass	old	0.35	0.7	0	4
Navigation	GPS watch	new	0.06	0.05	280	7
Water	Aluminium bottle	old	0.5	1	0	4
Water	Soft flasks	new	0.12	0.35	18	5
Poles	Wooden trekking poles	old	0.95	2	0	4
Poles	Carbon trekking poles	new	0.48	1	95	6
Optional	Film camera	old	0.75	1.6	0	3
Optional	Power bank	new	0.3	0.2	25	3
Optional	Paperback	old	0.45	0.6	0	2
Optional	Extra snack supplies	new	0.9	1.4	20	3
Optional	Binoculars	old	0.6	0.8	0	2
Optional	Bluetooth speaker	new	0.35	0.5	30	1
Optional	Playing cards	old	0.1	0.05	3	2
Optional	Sunscreen	new	0.15	0.1	8	2
Optional	Pony Pia (lucky charm)	new	0.05	0.1	6	8

Table 4: Equipment. Necessary groups: Sleeping, Light, First aid, Navigation, Water (exactly one item each). Origin: old = grandparents' attic (price 0), new = shop.

Remark: Pony Pia is a small plush horse that your friend Pia gave you for the trip. It does not help in any practical way, but it is said to bring luck – whether she is worth the space in your backpack is up to you.

You have three goals that pull in different directions: a high **benefit**, a low **weight** and a low **price**. They are measured in different units, so you have to find a way to combine them into one optimization problem. The way to do so is up to you.

a) Formulate a linear mixed-integer model for the equipment and implement it in GAMS. Decide how to deal with the three goals and **justify your approach** (e.g. by what kind of hiker you are); if you introduce weights or limits, justify their values. Which items do you pack? State the benefit, the total weight, the volume and the costs.

b) Solve the model for at least two further, clearly different settings of your approach (e.g. an "ultralight hiker" and a "budget hiker"). Which decisions change? Are there items that you pack in any case? Why?

(2) What to wear?

Now you decide which clothing to take and what to wear on each of the four stages. The clothing is listed in Table 5. There are five categories: top, mid layer, shell, legs and head. In addition, you can choose *no mid layer* and *no shell* (weight, volume and price 0), e.g. on a warm day.

Category	Item	Origin	Weight	Volume	Price	Properties (0–3)				
			(kg)	(l)	(€)	breath.	wind	rain	warm	sun
Top	Cotton shirt	old	0.25	1.2	0	1	0	0	1	0
Top	Merino shirt	new	0.15	0.7	60	3	0	0	2	0
Top	Synthetic shirt	new	0.12	0.6	35	3	0	0	1	0
Mid layer	Wool sweater	old	0.7	5	0	1	1	0	3	0
Mid layer	Fleece jacket	new	0.4	2.5	70	3	0	0	2	0
Mid layer	Light down jacket	new	0.3	1.5	150	1	2	1	3	0
Shell	Rain poncho	old	0.5	1.5	0	0	1	2	0	0
Shell	Hardshell jacket	new	0.45	2	200	2	3	3	0	0
Shell	Windbreaker	new	0.15	0.6	60	2	3	1	0	0
Legs	Corduroy trousers	old	0.8	3	0	0	2	1	2	0
Legs	Convertible pants	new	0.45	1.6	80	2	1	1	1	0
Legs	Shorts	new	0.2	0.7	40	3	0	0	0	0
Head	Straw hat	old	0.1	1.5	0	1	0	0	0	3
Head	Beanie	new	0.08	0.3	20	1	1	0	3	0
Head	Cap	new	0.1	0.5	15	2	0	0	0	2

Table 5: Clothing. Properties: breathability (breath.), protection against wind, rain and cold (warm), and protection against sun. A higher value means a better property.

Rules.

- On each stage you wear **exactly one item per category** (including the options “none” for mid layer and shell). You can only wear what is in your backpack, and for each category at least one real item is packed.
- In total (equipment and clothing), the backpack may contain at most **30 litres**, **9 kg** and cost at most **500 €**.
- There is no laundry: wearing the same item on more than one stage is allowed, but it becomes unpleasant. For each item that you wear on a stage *and have already worn on an earlier stage*, a comfort penalty applies.

How good is a piece of clothing on a stage? Not every property matters on every stage: warmth is great on a cold ridge and annoying on a hot climb. For every stage d and every property c (breathability, wind, rain, warm, sun) you therefore choose an **importance** $W_{d,c}$ between -3 and 3: 3 means “very useful on this stage”, 0 means “does not matter”, and a negative value means “disturbing”. Derive the importance from the forecast and the rules above and **justify your choice**. The **suitability** of an item i on stage d is then the sum over all properties of importance times property value ($A_{i,c}$ from Table 5):

$$SU_{i,d} = \sum_{c} W_{d,c} \cdot A_{i,c}.$$

Example: if breathability has importance 3 and warmth -2, a synthetic shirt (breathability 3, warm 1) scores $3 \cdot 3 - 2 \cdot 1 = 7$.

How tired are you after a stage? You start with a stamina of $S_0 = 100$ points. Each stage costs you its effort E_d (see above). The scores of the items you wear make the stage more pleasant and give stamina back: the *comfort* of a stage is the sum of the scores of the items you wear, minus $\kappa = 2$ for each item that you already wore on an earlier stage (no laundry!). Only a fifth of the comfort counts ($\gamma = 0.2$). During the night in the hut you recover $R = 15$ points, and with Pony Pia in your backpack $\rho = 3$ points more. In formulas (Pia $= 1$ if Pia is packed, otherwise 0):

$$s_d = s_{d-1} - E_d + \gamma \cdot c_d + R + \rho \cdot \text{Pia}, \qquad c_d = \sum_i SU_{i,d} \cdot \text{wear}_{i,d} - \kappa \cdot (\text{number of repeated items on stage } d).$$

with $\text{wear}_{i,d}$ being a binary variable indicating whether item i is worn on stage d. The objective is to maximize the **average stamina** over the four stages.

c) Extend your model: take the equipment of Part 1 as given (fixed!) and optimize the clothing. Choose and justify the importance weights $W_{d,c}$. What do you pack and what do you wear on each stage? State the stamina at the end of each stage and the average.

d) How does the result change if the penalty for repeated wearing is removed ($\kappa = 0$) or increased ($\kappa = 5$)? Which items are worn repeatedly in the different cases?

e) The budget is varied (330, 500, 650 €), and in a further scenario Pony Pia stays at home ($\rho = 0$). Summarize the results in a table (average stamina, items bought) and explain what you observe. Which additional items are bought when the budget grows?

Hint. Model the "already worn on an earlier stage" with auxiliary binary variables, e.g. $used_{i,d} \geq wear_{i,d-1}$ and $used_{i,d} \geq used_{i,d-1}$. The repeated wearing is then $rep_{i,d} \geq wear_{i,d} + used_{i,d} - 1$.

3 Are you (b)ready for it?



Figure 5: Tasty bread.

Fresh sourdough bread in the house – a new bakery was opened a couple of months ago. After gaining some first experience, the owner now wants to make some money. Since the owner knows you from earlier days and knows that you are an engineer, you are asked for help.

The bakery offers five different products, all homemade, of course. The (top-secret) recipes, the selling prices and the requirements of the production are given in Table 6. All quantities in this task refer to **one week with 6 working days**; products can be produced in arbitrary (also non-integer) quantities.

	Product	Recipe (kg per piece)				Price	Oven	Labor	Max. demand
		wheat	rye	butter	other	(€)	(min)	(min)	(pieces/day)
S1	Country sourdough	0.5	0.1	0	0.02	4.20	2	2	300
S2	Rye sourdough	0.25	0.35	0	0.02	4.50	2.2	2.5	150
S3	Seeded spelt sourdough	0.45	0	0	0.08	4.80	2	2.8	120
P1	Butter croissant	0.06	0	0.03	0	1.60	0.5	3	250
P2	Cinnamon roll	0.07	0	0.02	0.02	1.90	0.6	2.2	200

Table 6: Products. “Oven” and “Labor” denote the oven time and the working time that is needed per piece.

The ingredients are bought at the costs of Table 7 (no quantity limits). The oven capacity of the bakery is

900 minutes per day, and the available working time (all employees together) is **1350 minutes per day**. The demand per product is limited by the numbers given in Table 6.

Ingredient	Costs (€/kg)
Wheat flour	0.80
Rye flour	0.90
Butter	8.00
Other (seeds, sugar, cinnamon, . . .)	3.00

Table 7: Ingredient costs

You are asked to decide how much of each product the bakery should produce to maximize the **profit** by formulating a **linear** (mixed-integer) optimization model, implementing it in GAMS and providing a definite answer to the following tasks. The tasks build on each other.

Additional remark: Operating costs of the infrastructure (rent, energy, . . .) and the wages of the existing staff are neglected.

(1) Basic model

a) Formulate and solve a model that maximizes the weekly profit (revenue minus ingredient costs). How much of each product should be produced? State the profit.

b) Which of the products are not produced, although each of them has a positive profit per piece? Explain why. Which resource limits the production? How much would the weekly profit increase if one more minute of oven time or one more minute of working time were available (marginal values)?

(2) Supply chain of flour

So far, the flour was bought at a fixed price. Now, the bakery buys its flour directly from local mills. Three kinds of flour are available: standard wheat flour, premium wheat flour and standard rye flour. The mills offer different kinds at different prices and up to a maximum amount per week (Table 8). Butter and other ingredients are still bought at the prices of Table 7.

The wheat part of the recipes of the products S3, P1 and P2 has to be made of **premium** wheat flour. The other products may use any wheat flour. Rye flour is only available as standard flour.

Mill	Distance (km)	Std. wheat	Premium wheat	Std. rye
M1	12	0.82 / 1500	–	0.95 / 600
M2	35	0.78 / 1500	1.05 / 1200	–
M3	60	–	0.88 / 1000	0.80 / 800
M4	25	–	1.00 / 600	0.84 / 700

Table 8: Mills: distance by road from the bakery, and cost (€/kg) / maximum amount (kg/week) per kind of flour. "–": not offered.

The flour is transported to the bakery by trucks. Two types of trucks are available (Table 9). Each trip goes from the bakery to **one** mill and back (the length of a trip is therefore twice the distance of the mill) and transports flour from this mill only. A trip costs the fixed cost plus the cost per kilometre. A truck can

carry up to its capacity (also fewer), and the number of trips is not limited (at most 12 per mill and truck type, to keep your model small). The electric truck cannot drive more than its range in one trip.

Truck	Capacity (kg)	Fixed cost/trip (€)	Cost (€/km)	Range (km)
Conventional (diesel)	500	25	1.10	no limit
Battery-electric	300	40	0.60	100

Table 9: Trucks

If at least **70 %** of all trips in a week are done with electric trucks, the city government pays a subsidy, which corresponds to **240 € per week**.

c) Extend the model of (1): Find out which production plan, which purchase of flour from which mill, and which transport schedule maximize the weekly profit (including the subsidy). Summarize the result: profit, production, flour bought per mill and kind, trips per mill and truck type.

d) Is it worth fulfilling the 70 % rule? Compare the result with the case where the rule (and the subsidy) is ignored. From which subsidy per week on does it pay off for the bakery to use mostly electric trucks?

e) Which mills are not used at all, and why? (Short explanation using the data.)

(3) Investment

Assume that all your suggestions work out and the business flourishes. The owner now considers increasing the capacity. To be confident in the decision, you are asked to check this mathematically using your model. The following investments are possible:

Option	Effect per unit	Costs per unit	Maximum
Additional oven	+1500 oven minutes per week	15 000 € (one-off)	4
Additional employee	+2400 working minutes per week	900 € per week (wage)	3

The investment in an oven has to pay off within **104 weeks**; for the weekly planning, use the investment divided by 104 as weekly costs.

f) Extend your model of (2) by these options (whole numbers of ovens and employees). How many ovens and employees should be added? State the weekly profit (after the costs of the investments) and compare it with the result of (2).

(4) Interpretation

g) In (1), one oven minute has a marginal value of more than 1.50 €. One additional oven (1500 min) is therefore worth more than 2000 € per week, but it costs only about 144 € per week. Why does the model of (3) still not buy the maximum of four ovens?

4 Thermal Spa: Warm, warmer, cheapest



Figure 6: *Blue Lagoon* in Iceland, a famous thermal spa that does not require artifical heating :)

A thermal spa in the Alps is open all year round, and heating it is one of its biggest costs. In winter, three things have to be kept warm at the same time: the **pools**, the **air in the spa hall** (ventilation) and the **hot water** for showers. The owner asks you how this heat can be supplied as cheaply as possible. You look at one typical **week in November** (168 hours, one time step per hour).

What do these things do?

Background. Gas boiler: burns gas and produces heat. About 90 % of the energy of the gas ends up as heat. Gas costs 0.10 €/kWh (including grid fees and CO_2 costs).

Solar collectors: produce heat for free, but only when the sun is shining. Heat that is not used is lost.

Heat pump: works like a fridge running backwards. It needs electricity, but delivers *several* times as much heat from the outside air, e.g. 3 kWh of heat for 1 kWh of electricity. This factor is called *COP*. The colder it is outside, the lower the COP.

Heat storage: a large, insulated hot-water tank (like a thermos). You can charge it when heat is cheap or when there is spare capacity, and use it later. About 0.5 % of its content is lost every hour.

Units: power in kW, energy in kWh. Running 1 kW for one hour is 1 kWh. Since the time steps are one hour long, a power of 100 kW in one time step means 100 kWh of energy.

Data

The file `Thermalbad_Daten.inc` contains the data of the week for each hour t (Figure 7): the heat demand of the pools, of the ventilation and of the hot water (kW), the available solar heat (kW), the outside temperature (°C), the COP of the heat pump and **two** electricity price profiles (€/kWh): `pEl` (today) and `pElRen` (a scenario, see part (4)). Include the file in your GAMS model with the line `$include Thermalbad_Daten.inc`. It defines the set `t` (hours) and the table `pData(t,k)`.

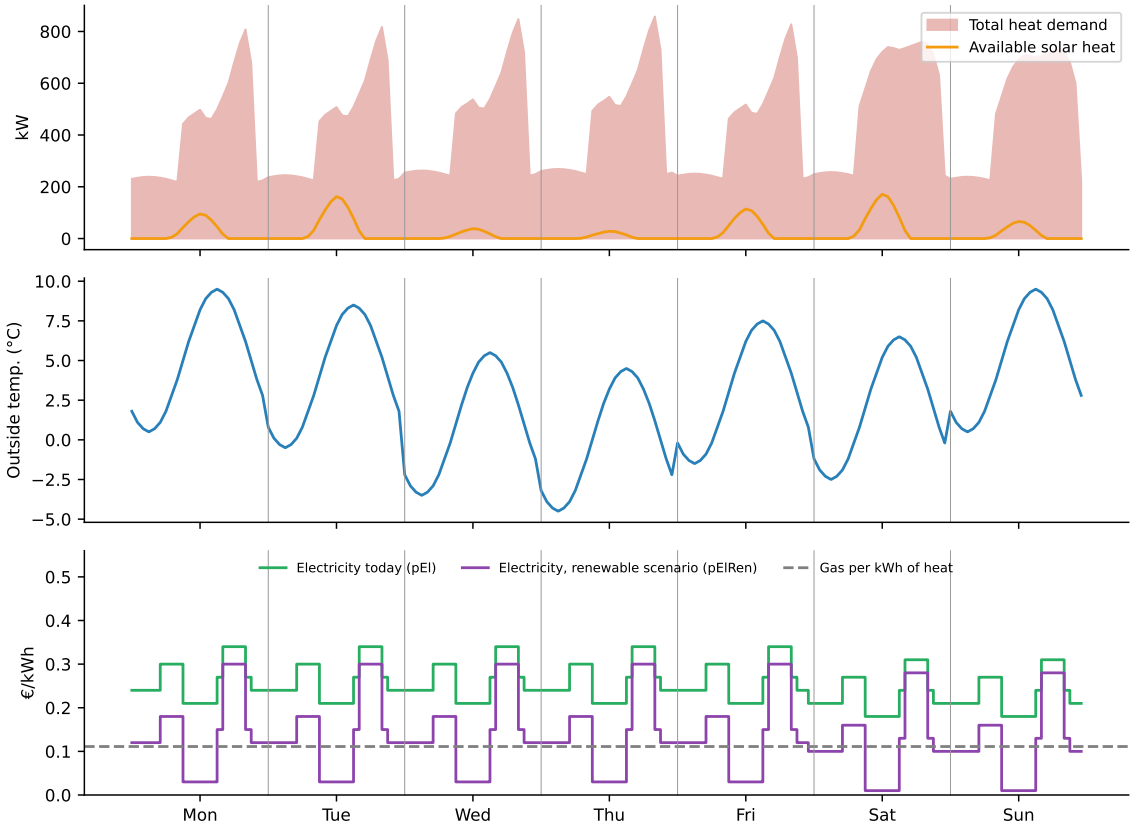


Figure 7: Data of the week: heat demand and available solar heat, outside temperature, and the two electricity price profiles (the dashed line shows the cost of one kWh of heat from the gas boiler)

Technical data. The gas boiler can deliver up to 900 kW of heat. The heat pump has a heat output of up to 450 kW. The heat storage can hold 2000 kWh and can be charged and discharged with at most 600 kW. It is half full in the beginning of the week and has to be at least half full again at the end. For parts (1) to (3), use the electricity price `pEl`.

Additional remark: Maintenance, personnel and the costs of the existing plants are neglected. All demand has to be covered in every hour.

Formulate a **linear** (mixed-integer) optimization model, implement it in GAMS and answer the questions. The tasks build on each other.

(1) Heat supply

a) Formulate and solve a model that covers the heat demand in every hour at minimal costs with the **gas boiler, the solar collectors and the heat pump** (without storage). State the costs of the week.

b) How much money per week does the heat pump save compared to supplying everything with boiler and solar?

(2) Heat storage

c) Extend the model by the heat storage. State the costs of the week and the savings compared to a). At which times is the storage charged and discharged? Explain why this saves money.

d) What changes if the electricity price is the same in all hours (e.g. 0.25 €/kWh)? What do you conclude about the reasons why the storage is useful?

(3) Investment

The owner considers extending the plant. The following options are available:

Option	Effect per unit	Costs per unit	Maximum
Additional heat pump	+150 kW heat output	200 € per week	4
Additional heat storage	+1000 kWh capacity	100 € per week	3

The costs per unit are already converted to one week (including depreciation).

e) Extend your model of (2) by these options (whole numbers of units). How many units should be bought? State the total costs of the week (operation plus investment) and compare them with c).

(4) Comparing two scenarios

Background. So far, you optimized the spa for *today's* electricity prices (`pEl`). The data file also contains a second profile, `pElRen`, which describes a possible *future* week: the electricity price follows the amount of wind and solar power. At noon the price is very low, in the evening it is high. In this part you compare the two scenarios, i.e. the same spa with the two price profiles.

f) Solve the models of (2) and (3) again, now with the electricity price `pElRen` instead of `pEl`. State the costs of the week without investment and the optimal investment with the resulting costs. Compare with c) and e): How do the costs, the use of the boiler, and the optimal investment change?

g) Explain the differences. When is the storage charged and discharged in the new scenario, and why? Give a short recommendation for the owner: what should be bought, and what does the result mean for the role of heat pumps and storage in a renewable energy system? Where are the limits of your model (which assumptions should the owner check)?

Hint. In GAMS, the previous hour of `t` can be addressed with `t-1`. For the first hour, which has no previous hour, you need a separate treatment (e.g. with a condition like `$(ord(t) > 1)`).

5 Crop Circles: Rotate, rotate, rotate



Figure 8: Rice fields near Toyooka, Japan. An example of a monoculture that is not as problematic because the rice is grown in water.

A farm in eastern Styria cultivates **six fields** with a total area of 60 ha. Every year, exactly one crop grows on each field. The farmer is planning the **next six years** and asks you which crop should grow on which field in which year. The aim is the highest possible **total profit** over the six years.

Why not simply grow the best crop every year?

Background. Farmers *rotate* their crops (crop rotation). If the same crop grows on the same field year after year, diseases and pests build up in the soil, and the soil is drained of the same nutrients again and again. Different crops also need work at different times of the year. **Legumes** (here: soybeans and clover-grass) are special: they take nitrogen from the air and leave it in the soil, so the crop after them needs less fertilizer. This is called the *nitrogen credit*.

Oil pumpkin is a Styrian specialty: its seeds are pressed into pumpkin seed oil. **Clover-grass** is sold as hay. The costs in the table cover seed, fertilizer, plant protection and machines.

Data

The seven crops are given in Table 10. The numbers are rounded and simplified.

Crop	Yield (t/ha)	Price (€/t)	Costs (€/ha)	Work (h/ha)	
				spring	autumn
Wheat	7.0	210	700	3	5
Maize	10.5	185	1000	5	8
Soybeans *	3.0	450	450	4	4
Oil pumpkin	0.70	3300	1000	8	25
Rape	3.6	450	750	3	5
Potatoes	38	150	3700	22	30
Clover-grass *	9.0	120	600	3	6

* legume. The yield of oil pumpkin refers to dry seeds, the yield of clover-grass to hay.

Table 10: Crops

The fields differ in size and soil. The **yield factor** in Table 11 tells how much of the yield of Table 10 the field delivers for each crop (e.g. 0.90 means 10 % less). A factor of **0** means that the crop is not possible on this field (the soil of F2 is too heavy for potatoes, F4 is too steep).

Field	Area (ha)	Wheat	Maize	Soy	Pumpkin	Rape	Potatoes	Clover	Crop last year
F1	12	1.00	1.00	1.00	1.00	1.00	1.00	1.00	Wheat
F2	8	1.05	0.95	0.85	0.90	1.05	0	1.00	Maize
F3	15	1.10	1.10	1.05	1.10	1.05	1.10	1.00	Soy
F4	10	0.95	0.90	1.00	1.00	0.90	0	1.10	Rape
F5	6	0.90	0.85	0.90	0.95	0.85	1.05	0.90	Clover
F6	9	1.00	1.05	0.90	0.80	1.00	0.90	1.10	Pumpkin

Table 11: Fields and yield factors

Enter these data in your GAMS model as sets, parameters and tables. The **farm's own work force** is limited: in each year there are at most **450 working hours in spring** and **800 working hours in autumn** available for all fields together.

Formulate a **linear** (mixed-integer) optimization model, implement it in GAMS and answer the questions. The tasks build on each other.

(1) A first plan

a) Find the plan with the highest total profit over six years under these rules: on every field exactly one crop is grown per year; the same crop must not be grown on the same field two years in a row (this includes last year's crop); a crop with yield factor 0 is not possible; one crop may cover at most **40 %** of the total area (market risk); the working hours in spring and autumn are limited. State the profit and show the plan as a table (fields $\times$ years).

b) Which of the limits (area share, spring work, autumn work) restricts the farm, and which one hardly matters? Estimate the value of 100 additional working hours in spring and of 100 additional hours in autumn (over six years).

(2) Proper crop rotation

The plan of (1) swings back and forth between two crops on a field. A good rotation needs more rules. Add the following to the model:

- A legume must not follow a legume (they suffer from the same diseases). This also holds for last year's crop.
- **Rotation breaks:** on the same field, potatoes and rape may be grown at most once in any **four** consecutive years, and oil pumpkin at most once in any **three** consecutive years. In every **three** consecutive years, at least one legume must grow on each field. (Only the six planning years are considered.)
- **Nitrogen credit:** if the previous crop on a field was a legume, the farmer saves fertilizer worth 60 €/ha (after soybeans) or 90 €/ha (after clover-grass). This also applies in year 1 for last year's crop.

c) Solve the extended model and show the plan. By how much does the profit change compared to a)?

Hint. `alias` indices and `ord()` may be useful.

(3) Investment and subsidy

Potatoes need a **harvester** that the farm does not own yet. Over the six years, buying it costs **20 000 €** (all-in, one decision). Without the harvester, no potatoes can be grown. In addition, there is a **biodiversity subsidy**: in every year in which at least **6 different crops** are grown on the farm, the farmer receives 55 € for *each* hectare of the whole farm (60 ha).

d) Extend the model of (2) by the harvester and the subsidy. Should the farmer buy the harvester? In which years is the subsidy received? State the total profit (including investment and subsidy) and show the plan.

e) Compare with the situation without harvester. Up to which price would the harvester still pay off? (Try out some prices.) Briefly explain how the subsidy changes the plan.

6 Plug and Sail: Electric Ferries



Figure 9: Ferry approaching a pier.

In the Seabird Archipelago, people, cars and goods cross the fjords by **car ferry**. The regional authority operates **10 ferry routes** between 19 piers (Figure 10). Today, all ferries are driven by **diesel engines**. The authority wants to know which routes should be converted to **battery-electric** operation, where **chargers** have to be built at the piers, and what this costs and saves. You are the optimization expert.

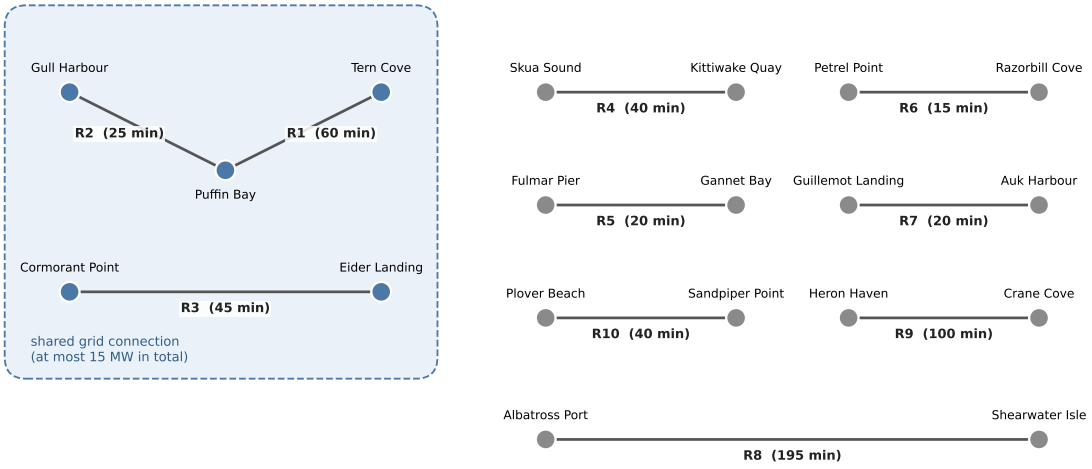


Figure 10: The ferry network: 19 piers and 10 routes with their crossing times. The five blue piers are supplied by one common grid connection, therefore the max. charging power of all chargers must not exceed 15 MW.

What do these things do?

Background. Energy and power: Energy is measured in MWh, power in MW. Running at 1 MW for one hour uses 1 MWh; 5 MW for 12 minutes (0.2 h) is 1 MWh. A ferry needs a *sailing power* (in MW) to move, so one crossing needs *power* $\times$ *crossing time* of energy.

Battery ferry: The diesel engine is replaced by electric motors and a large battery. The battery is recharged at the pier while the ferry waits for the next departure (the *layover time*). A **charger** at the pier delivers the energy with a certain charging power: charging energy $=$ power $\times$ layover time. Chargers need a grid connection, so they are expensive.

Diesel: A diesel ferry burns 270 litres of diesel for every MWh of energy it needs for sailing. Every litre emits 2.68 kg of CO_2, and CO_2 has a price. An electric ferry emits nothing locally; but when charging, 10 % of the electricity is lost (charging efficiency 90 %).

Annual costs: An investment is used for many years. To compare it with yearly savings, it is converted into *annual costs*: yearly cost $=$ annuity factor $\times$ investment. A subsidy pays part of the investment.

Data

The routes are given in Table 12. Each route is sailed by one or two ferries (all ferries of a route are identical) in both directions. *Departures* are per day and direction (including all ferries); the route runs on the given number of days per year (R9 is a summer route only).

Route	Piers	Crossing (min)	Departures per day	Days per year	Ferries	Sailing power (MW)	Layover (min)
R1	Puffin Bay – Tern Cove	60	14	365	2	2.1	25
R2	Puffin Bay – Gull Harbour	25	24	365	2	1.8	20
R3	Cormorant Point – Eider Landing	45	9	365	1	1.6	25
R4	Skua Sound – Kittiwake Quay	40	9	365	1	0.9	40
R5	Fulmar Pier – Gannet Bay	20	15	365	1	1.0	30
R6	Petrel Point – Razorbill Cove	15	6	365	1	0.5	40
R7	Guillemot Landing – Auk Harbour	20	7	365	1	0.5	40
R8	Albatross Port – Shearwater Isle	195	2	365	2	6.0	60
R9	Heron Haven – Crane Cove	100	2.5	150	1	2.0	120
R10	Plover Beach – Sandpiper Point	40	4	365	1	0.7	120

Table 12: Routes (departures are per day and direction; 2.5 means 2 and 3 on alternating days)

The other data are given in Table 13. All amounts of money are in **CU** (currency units, MCU $=$ million CU).

Diesel consumption	270 l/MWh	Conversion of one ferry to electric	8 MCU
Diesel price	12 CU/l	Battery	4 MCU per MWh
CO_2 emission of diesel	2.68 kg/l	Charging pier (fixed cost)	30 MCU
CO_2 price	1000 CU/t	Charger	5 MCU per MW
Electricity price	1000 CU/MWh	Subsidy (share of the investment)	25 %
Charging efficiency	90 %	Annuity factor	10.3 % per year

Table 13: Prices, costs and factors

Planning rules. (i) **Battery:** Only half of a battery may be used (typically such a battery is operated between 20 % and 80 % state of charge (SoC); here we model this by assuming that only half of the battery capacity is available for use). A ferry therefore needs a battery of *twice* the energy it uses between two charging stops. If it is charged at **both** piers of its route, this is one crossing; if it is charged at only **one** pier, it sails there and back, i.e. two crossings. The largest battery that can be built is **8 MWh**.

(ii) **Charger power:** A pier where a ferry is charged needs a charger with at least *energy to be recharged* / *layover time*. If several routes use the same pier, their ferries charge one after the other, so the charger only has to cover the *highest* need of these routes (but is built and paid only once).

(iii) **Grid:** The five blue piers (Puffin Bay, Tern Cove, Gull Harbour, Cormorant Point, Eider Landing) are in close proximity and therefore share one connection to the transmission grid (vgl. "ein gemeinsames Umspannwerk"). To be on the safe side, assume that all chargers there may run at the same time: the *sum* of the charger powers at these piers must not exceed **15 MW**.

(iv) A route can be charged at one or at both of its piers. An electric route needs at least one charger.

Formulate a **linear** (mixed-integer) optimization model, implement it in GAMS and answer the questions. The tasks build on each other. Treat the ten routes as one network: decisions at piers that belong to two routes (here: Puffin Bay) concern both routes.

(1) A single route

Look at route R1 (Puffin Bay – Tern Cove) alone and answer the following questions by doing some calculations – this is meant to be a guided exercise in order to understand the data and the problem. It is not meant to be solved using *GAMS*.

a) How much energy does one crossing need, and how much per year (both directions, all departures)? How much do you save per year when the diesel ferries are replaced (diesel and CO_2 costs minus electricity costs)?

b) Which charging variants (one pier, both piers) are technically possible? State battery size per ferry and charger power per pier.

c) Calculate the investment for a possible variant (ferries, batteries, chargers), subtract the subsidy and convert it into annual costs. What is the annual net benefit, and after how many years is the investment (after subsidy) paid back by the savings?

(2) The whole network

The authority wants the **lowest net annual cost**: annual investment costs (after subsidy) minus annual savings. Decide for every route whether it is electrified, at which piers chargers are built (one or both ends of a route), and how large batteries and chargers are.

d) Formulate and solve the model. Which routes are electrified, which piers get a charger, which routes are charged at one and which at both ends? State the net annual cost (benefit) and the share of the CO_2 emissions of the diesel network that is avoided.

e) Some routes are not electrified. Find out for each of them *why not*. Do this with your model, e.g. by forcing the route into the solution or by relaxing one rule.

Hint. You need a binary variable for each route (electrified or not), one for each pier (charger or not), and one for each route end (ferry charged at this pier or not). Sums of these variables tell you whether a route is charged at one or at both ends. Battery size and charger power then follow from the rules above; where a rule only holds “if the charger exists”, a big-M constant helps.

(3) Grid and climate target

The authority can **reinforce the grid** at the five blue piers: for 40 MCU the limit of 15 MW rises by 6 MW.

f) Add this option (before subsidy; treat it like the other investments). Is it worth it? What changes in the solution?

g) The authority now sets a **climate target**: the emissions of the diesel network are to be reduced by at least 50 %, 55 %, 60 %, 62 %, 64 %, 65 % and 66 %, respectively (keep the reinforcement option). Solve the model for each target and state the net annual cost. What is the largest reduction that can be reached? Which route stops the authority from going further?

(4) Carbon price

The CO_2 price of 1000 CU/t is uncertain. Solve the model of (3) **without** climate target for CO_2 prices of 0, 500, 900, 1500, 2000, 2500 and 3000 CU/t.

h) Which routes are added at which price? Estimate, from your results, the CO_2 price from which the borderline route is electrified. Which routes are never electrified by prices alone?